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Dallas, Texas, USA

Quick-Start Installation Instructions

ComfortSense® 7000 Series Model No. L7742U

Touch Screen Programmable Thermostat

IMPORTANT! READ THIS FIRST!

Before beginning installation, note the type of equipment, number of stages, and any accessories being installed. The ComfortSense® L7742U thermostat is a 24VAC low-voltage thermostat and requires a common wire to the thermostat to operate. Observe the specifications and the DO's and DO NOT's that follow:

- Use 18 gauge thermostat wire or larger.
- Maximum load 1 Amp or less.

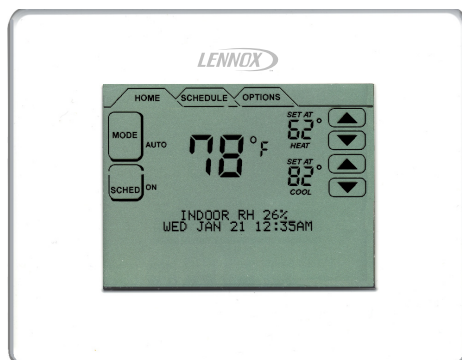
- Maximum wire run length 300' (one way).

DO

- Shut off all power to system before installing
- Read this entire document, noting which instructions pertain to your equipment and system requirements.
- Make sure that all wiring conforms to local and national building and electrical codes and ordinances.
- Use thermostat wire between the thermostat and the optional outdoor temperature sensor (may be separate wire pair or two wires of a multi-wire cable). No shielded or twisted-wire pair is needed.
- Consult the ComfortSense® 7000 Series Programming and Application Guide 506228-01.

DO NOT

- Install on voltages higher than 30VAC.
- Short (jumper) across terminals on the gas valve or at the system control to test installation. This will damage the thermostat and void the warranty. **Instead, use testing function in the thermostat shown on back page.**
- Install on outside walls or in direct sunlight
- Install for commercial applications unless used with approved split-system matches, and all wiring and load criteria is been met.



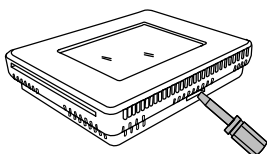
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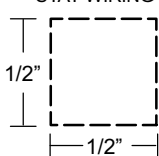
Install base in desired location

UNPACKAGE THE THERMOSTAT AND OPEN THE CASE WITH A FLAT-BLADE SCREWDRIVER. PLACE BETWEEN BASE AND UNIT AND TWIST TO SEPARATE UNIT FROM BASE.



1

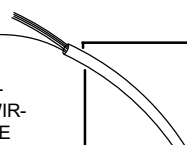
CUT OR DRILL A SMALL HOLE FOR THERMOSTAT WIRING



2

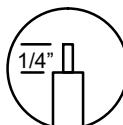
PULL ABOUT 3" OF THERMOSTAT WIRE THROUGH OPENING AND REMOVE OUTER THERMOSTAT WIRE JACKET.

THIS WILL HELP IN ROUTING THE THERMOSTAT WIRING TO THE PROPER THERMOSTAT TERMINALS



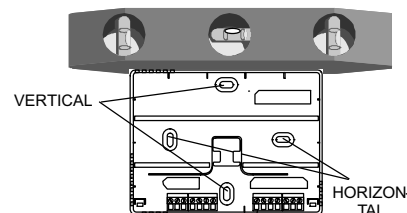
3

TRIM 1/4" INSULATION FROM END OF EACH WIRE



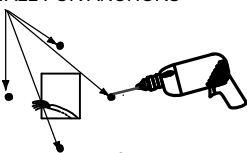
4

USE THE UNIT BASE AS A TEMPLATE AND A LEVEL TO MARK TWO MOUNTING HOLE LOCATIONS ON WALL. USE EITHER VERTICAL OR HORIZONTAL HOLES.



5

DRILL TWO 3/16" HOLES AT EITHER HORIZONTALLY OR VERTICALLY MARKED LOCATIONS ON WALL FOR ANCHORS



6

INSERT WALL ANCHORS IN HOLES.



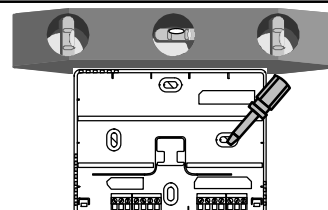
7

TAP DOWN THE ANCHORS UNTIL FLUSH WITH WALL.



8

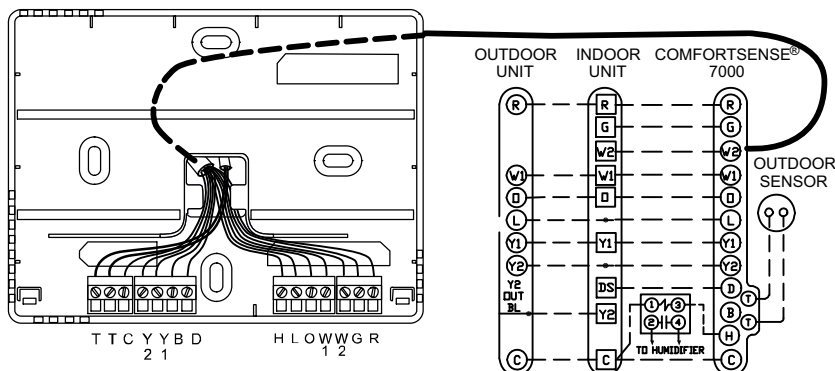
(USE A LEVEL) ALIGN BASE OVER PLASTIC ANCHORS; SECURE TO WALL USING PROVIDED SCREWS.



9

Connect wiring, seal hole, and setup thermostat

1. Connect wiring between thermostat, indoor unit, and outdoor unit similar to that shown below. See unit-specific wiring diagrams on back page.
2. Connect the outdoor sensor (if used) to the 'T' connections on the thermostat (wire run should not exceed 300' [91m]).
3. Seal the hole in the wall with a suitable material to prevent drafts from entering the thermostat case.
4. Configure thermostat and equipment for system type, program the thermostat, and test system (see minimum configuration guide on back page or *Programming and Application Guide*).



TERMINAL DESIGNATIONS

- T - Outdoor Temp. Sensor Connection 1 (Opt.)
- T - Outdoor Temp. Sensor Connection 2 (Opt.)
- C - Common 24 Vac
- Y2 - 2nd stage heating or cooling
- Y1 - 1st stage heating or cooling
- B - Heat active reversing valve
- D - Dehumidification relay (to DS terminal)
- H - Humidification relay (to Humidifier)
- L - System check (HVAC Fault Detected?)
- O - Cool active reversing valve
- W1 - 1st stage heating (non-heat pump or emergency heat) or 3rd stage heating (heat pump)
- W2 - 2nd stage heating (non-heat pump) or 4th stage (heat pump)
- G - Fan relay
- R - 24 Vac power

Wiring Diagrams

1 WHEN 2-STAGE HEAT IS USED, T'STAT IS CONFIGURED FOR MULTISTAGE HEAT, CONNECT T'STAT V2 TO AIR HANDLER V2 & REMOVE JUMPER BETWEEN R & V2.

2 CUT ON-BOARD LINK FROM DS TO R (V914 CLIPPABLE CONNECTION ON FURNACE/JUMPER ON AIR HANDLER) WHEN DEHUMIDIFICATION OR HARMONY CONTROL IS USED.

3 DO NOT CUT ON-BOARD LINK FROM R TO O (V951 CLIPPABLE CONNECTION ON FURNACE/JUMPER ON AIR HANDLER). IT IS USED FOR CONDENSING UNIT APPLICATIONS.

4 CUT ON-BOARD LINK FROM R TO O (V951 CLIPPABLE CONNECTION ON FURNACE/JUMPER ON AIR HANDLER).

5 REMOVE DS TO Y1 JUMPER WHEN DEHUMIDIFICATION D TERMINAL IS USED.

6 OUTDOOR SENSOR FOR OUTDOOR TEMP. DISPLAY, OPTIONAL HUMIDIFIER DEF. POINT CONTROL, HEAT PUMP BALANCE POINTS & DUAL FUEL BALANCE POINTS.

7 RED WIRE TO OUTDOOR CONDENSING UNIT USED ONLY FOR UNITS WITH LSM.

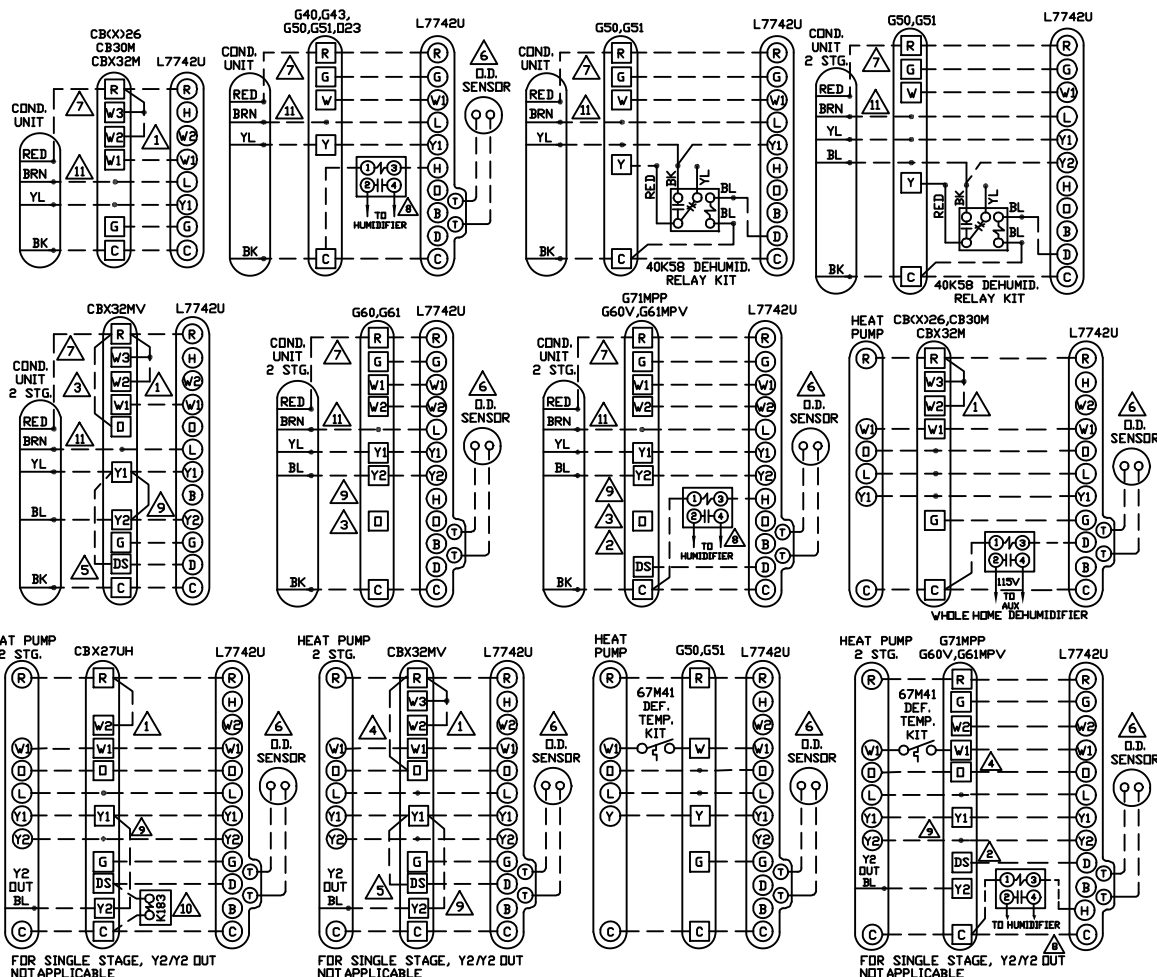
8 FOR HUMIDIFIER, 48G95 (OR EQUIVALENT) ISOLATION RELAY REQUIRED (COIL 24VAC, 5VA MAX.). CAN BE USED WITH ALL T'STAT APPLICATIONS.

9 FOR 2-STAGE CUT ON-BOARD LINK FROM Y1 TO Y2 (V915 CLIPPABLE CONNECTION ON FURNACE/JUMPER ON AIR HANDLER).

10 KI83 DEHUMIDIFICATION RELAY KIT (26W62) REQUIRED WHEN USING D TO DS TERMINALS WITH CBX027 UNITS.

11 BROWN WIRE USED ONLY ON COND. UNITS WITH LSM.

NOTE: NOT ALL UNUSED TERMINALS ARE SHOWN ON T'STAT DIAGRAMS.



Configure and test thermostat

Press **OPTIONS** tab, then use the down arrow to select **INSTALLER SETTINGS**. Press **ENTER**; a note states "MUST BE SET BY QUALIFIED PERSON". Press **Enter** again to access installer settings.

INSTALLER SETTINGS

CLEAN ENTER

SYSTEM SETUP—With **SYSTEM SETUP** selected, press **enter**. Set thermostat for operation with a heat pump or non-heat pump. Set number of compressor stages. Set number of backup heat stages. **Defaults:** Non-heat Pump, Gas/Oil, 2 compressor stages, 2 indoor heat stages.

Use arrows to scroll through and select from the list; then press **ENTER**.

SYSTEM SETUP
NON HEAT PUMP
HEAT PUMP
COMPRESSOR STAGES
INDOOR HEAT STAGES

For **NON HEAT PUMP**, use arrows to select from the list; then press **SAVE**.

NON HEAT PUMP
NO HEAT
GAS/OIL
ELECTRIC

For **HEAT PUMP**, use arrows to select from the list; then press **SAVE**.

HEAT PUMP
NO BACKUP HEAT
DUAL FUEL (GAS/OIL)
ELECTRIC

For **COMPRESSOR STAGES**, use arrows to select 1 or 2 compressor stages; then press **SAVE**.

COMPRESSOR STAGES
2

For **INDOOR HEAT STAGES**, use arrows to select 1 or 2 indoor heat stages; then press **SAVE**.

INDOOR HEAT STAGES
2

SYSTEM TEST MODES—Press **OPTIONS** tab, go to **INSTALLER SETTINGS**, then press up arrow twice to select **SYSTEM TEST MODES**. Press **ENTER**; select **TEST OUTPUTS** and press **ENTER**.

SYSTEM TEST MODES

TEST OUTPUTS

Test all cooling, heating, Emergency Heating stages and FAN outputs. **SCROLL** arrows move through a list of all signals, Y1 ON, Y1 OFF, Y2 ON, Y2 OFF, etc. With a signal displayed, press **ENTER** to start the test, (e.g. Y1 ON selected, press **ENTER** brings on Y1; Y1 OFF selected, press **ENTER** shuts off Y1. **CANCEL**, pressed at anytime during tests will return the previous screen and also disable any test and puts the thermostat back into normal mode.

CAUTION! - In dual fuel system applications, do not turn on heat pump and furnace at the same time in system test mode.

All HVAC components can be tested to confirm the thermostat is sending signals to each unit.

NOTES - After 5 minutes without a test being initiated, the test mode is disabled and system goes back to the normal mode (i.e. **HOME** screen).

When in **SYSTEM TEST MODE**, the compressor minimum off timer is bypassed.

TEST OUTPUTS

OFF
Y1 ON
Y1 OFF
Y2 ON
Y2 OFF
W1 ON
W1 OFF
W2 ON
W2 OFF
FAN (G) ON
FAN (G) OFF
H ON
H OFF
D ON
D OFF
O ON (B OFF)
B ON (O OFF)

IMPORTANT!

The test mode only energizes the outputs in the order you turn them on. When Y1 or Y2 is turned ON, the FAN (G) output remains OFF. With the system connected, the compressor will be energized without the fan, unless you first set the FAN (G) to ON, then turn on the Y1 output.